

Part 2 of 2

Low-Power MCU Selection

Adding a low-power MCU to the main system reduces energy use during standby mode. But when selecting the one for your application, there are some areas of concern

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One major problem with most electrical products used in homes and offices is that they consume considerable amount of electrical power during standby mode. This power consumption can be controlled using MCUs and software techniques.

Part 1 of this article detailed low-power MCU parameters and transition to and recovery from low-power mode. Let's now proceed to dynamic power management using hardware and software techniques and key low-power features of commercially available MCUs.

Dynamic power management

Using ultra-low power MCU. Adding a general-purpose low-power MCU to the main system reduces energy use during standby mode, using specific firmware for power control.

The block diagram of a generic system comprising the main system block and the power management block is shown in Fig. 6. The main system block performs the primary function of the system, interfacing with a standby controller, wake-up sources and other devices.

The power management block is a general-purpose block independent of the main system. The brain of this subsystem is a low-power MCU typically used as standby controller. It receives inputs from the main IC controller and wake-up resources, and switches the

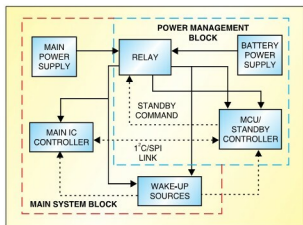


Fig. 6: The general system block diagram

power supply block (which supplies power to most of the main system) on and off through the relay. In active mode, the main IC controller and other devices are powered on. During standby mode, the low-power MCU and wake-up sources are powered by the auxiliary power supply. The default state after reset is the active mode.

The above hardware technique of dynamic power management can be implemented using general-purpose communication link (I²C/SPI) or non-maskable interrupt.

Any general-purpose input-output pin can be used for signaling, but there are specific advantages if the NMI pin is used for this purpose.

I²C/SPI communication link. This link supplies power to most of the main system. The main IC controller interfaces the low-power MCU using the I²C/SPI bus, where it is configured as the master and the low-power MCU is configured as the slave.

When the low-power MCU receives the standby command from the main IC controller, it puts the system in standby mode, switching off the relay connected to the fully-on power supply. During this phase, the low-power MCU enters sleep mode, waiting for an external input or timer event. Once any of these wake-up events

occurs, the low-power MCU exits the sleep mode and switches on the entire system through the relay.

The type and function of the wake-up event is communicated through an interrupt to the main IC controller to perform the related operation. This puts the system in active mode, and each normal function can run. During active mode, the main IC controller updates low-power MCU registers in order to program new timer events and to send the power-off command request.

Non-maskable interrupt. This is another mechanism to implement power management scheme for the MCU based on system on/off states by using its non-maskable interrupt (NMI) pin. Consider the block diagram shown in Fig. 7.

When the system reset is asserted, the RESET signal output from the main IC controller or reset controller goes low. This signals the MCU to en-